

Tirth Vasantkumar Patel

Calgary, AB, Canada | +1 (587) 377-6738 | tirth7411@gmail.com

LinkedIn | GitHub | tirthpatel.dev

Red Deer Polytechnic

Diploma in Software Development

Red Deer, AB, Canada

Recent Graduation: Sept 2025

- **Core Coursework:** Data Structures & Algorithms, Advanced Object-Oriented Programming, Database Management Systems, Distributed Architectures, Machine Learning Fundamentals.

Technical Skills

Languages: JavaScript (ES6+), Python, C# (.NET), TypeScript, SQL (PostgreSQL, MySQL), HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express, ASP.NET Core, Redis, Pandas, Scikit-Learn, NumPy

Cloud & Databases: Amazon S3, MongoDB, Docker, Git, Linux Environment, Postman, Jira, Confluence

Methodologies: Distributed Systems, RESTful API Design, System Architecture, OOP, CI/CD, Agile/Scrum

Projects

Real-Time Distributed Chat Application | *Node.js, Express, WebSocket, Redis, Docker, PostgreSQL*

- Architected a real-time bi-directional messaging service handling concurrent WebSocket connections with sub-100ms delivery latency.
- Implemented a Redis Pub/Sub layer to horizontally scale message broadcasting across multiple server instances dynamically.
- Designed an optimized relational schema in PostgreSQL using composite keys to index and fetch historical chat threads efficiently.

Algorithmic Pathfinding & Graph Visualizer | *TypeScript, React.js, HTML5 Canvas, CSS Grid*

- Engineered an interactive visualizer for classic graph traversal algorithms including Dijkstra's, A*, and Breadth-First Search (BFS).
- Optimized spatial search mechanics by implementing custom priority queues and min-heaps to minimize runtime execution complexity.
- Built an immutable state management system in TypeScript to dynamically render node state transitions without UI performance lag.

Scalable Full-Stack E-Commerce Platform | *React.js, Node.js, Express, MongoDB, REST APIs, Git*

- Architected a high-throughput, decoupled storefront utilizing a secure RESTful API ecosystem to seamlessly handle concurrent traffic.
- Built dynamic, reusable client-side views in React.js, implementing lazy loading to optimize asset rendering and state management.
- Structured a normalized MongoDB schema and deployed custom indexing strategies, reducing query latency by 35% under peak load tests.

Predictive Analytics Engine & Data Pipeline | *Python, Pandas, Scikit-Learn, NumPy, PostgreSQL*

- Engineered an end-to-end data processing pipeline to ingest, clean, and transform unstructured behavioral consumer datasets.
- Trained and cross-validated supervised machine learning models (Random Forests, Logistic Regression) to achieve a 92% retention forecast accuracy.
- Optimized data retrieval sequences by writing multi-table SQL queries, window functions, and aggregations for target feature matrices.

Containerized Microservice Log Aggregator | *Python, Docker, Amazon S3, FastAPI, Linux Shell Scripting*

- Developed a centralized log collection agent that scrapes, parses, and structures multi-format system logs from distributed containers.
- Orchestrated an asynchronous data pipeline using FastAPI to stream compressed log payloads to secure Amazon S3 storage buckets.
- Authored automated Bash scripts to handle container health-checks, resource rotation patterns, and system environment deployment.